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JES Focus Issue on Advances in Electrochemical Processes for Interconnect Fabrication in Integrated Circuits

Modeling of Dendrite Formation as a Consequence of Diffusion-Limited Electrodeposition

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Diffusion-controlled electrodeposition was investigated experimentally and by finite element method simulations of the concentration profile on microstructured band electrode arrays (MEA) and compared to existing theoretical approaches. The simulations revealed the establishment of a diffusion layer significantly larger than anticipated according to existing models for individual bands. The electrochemical depositions were simulated by extending the finite element method to allow the modeling of changing electrode geometries during growth by use of automated remeshing steps. As an ideal case of electroplating, thin films of copper electrodeposited onto MEA of gold on SiO_2/Si wafers were studied as a benchmark system. The electrodeposition of ZnO on MEA following initial reduction of O_2 served as an example of a multi-step reaction. The simulations considered an increasing depletion of the reactant (O_2 in the case of ZnO, Cu^{2+} in the case of Cu) toward the electrode bands and film growth limited by mass transport. The growth of experimentally observed dendrites and even their detailed size distribution over the whole array was precisely predicted by the simulations. An extended understanding of diffusion-controlled growth could be achieved and used to study the temporal development of electrochemical depositions, applicable also in complex geometries.

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The formation of dendrites is a well-known phenomenon in electrochemical processes. It has been observed, e.g., for the deposition of copper,[1] metallic lithium[2–10] or zinc oxide.[11,12] Whereas solutions were found already for the deposition of copper and metallic lithium, for zinc oxide no technique exists to control or at least predict the growth of dendrites. In the field of electrochemical copper deposition on microstructures Moffat et al. developed impressive ways to suppress preferred growth on edges to fill trenches and vias.[13–23] Formation of dendrites is not limited to liquid electrolytes but was observed in solid electrolytes, too.[24] Due to the danger of short circuits through metal dendrites and subsequent detrimental steps possible, the prevention of dendrite formation in general is a crucial part in the development of, e.g., batteries.[25,26] Also, in the deposition of tin-compounds for solder bumps or as interconnect materials for integrated circuit packaging the suppression of dendrite formation is still a challenging task.[27–29] On the other hand, whenever a large surface area is beneficial, a well-directed formation of dendrites could be of interest to improve the performance of electrodes.[30–32] Several theoretical descriptions and finite element method (FEM)-based simulations of concentration profiles and resulting currents during reduction and/or oxidation processes have been well developed.[33–43] In the electrodeposition of copper, e. g., uniform concentration distributions of additives even in complex structures have been aimed at and supported by numerical simulations on electrode arrays.[44–46] An influence of individual electrode spots on the distribution of local concentration profiles and, hence, on a geometrical dependence of the growth rate was indicated. Independently, the fractal shape of dendrites could be described by the diffusion-limited aggregation method (DLA) developed by Witten and Sander 1981.[47–49] To the best of our knowledge, however, there is still a lack of an adequate and flexible way to predict the formation and the shape of growing dendrites in a given electrode geometry especially on a real time scale and for reaction times sufficiently long to provide insight into "real-world technical problems". In turn, concentration profiles at larger dendrites could deliver interesting data to discuss diffusion-controlled processes.

By use of the commercially available finite element solver Fluent 16.2 (ANSYS) and the meshing tool ICEM CFD 14.0 we developed a routine to allow large mesh displacements with moving boundary conditions. Such approach is needed to allow modeling of dendrite growth under continuously changing geometric conditions. The routine is used to predict diffusion-controlled growth in the electrochemical deposition of porous zinc oxide (ZnO) or, as a model system, copper (Cu) as active electrode materials.

The electrochemical deposition of ZnO from an oxygen-containing aqueous electrolyte containing eosinY as structure-directing agent and electrocatalyst was found to be limited by the diffusion of oxygen toward the electrodes.[12] The underlying reactions can by expressed by 1 and 2 and summarized by 3:[50]

$$O_2 + 2H_2O + 4e^- \rightarrow 4OH^- \quad [1]$$

$$Zn^{2+} + 2OH^- \rightarrow Zn(OH)_2 \rightarrow ZnO{+}H_2O \quad [2]$$

$$Zn^{2+} + 0.5O_2 + 2e^- \rightarrow ZnO, \quad [3]$$

whereas the copper deposition can be described by

$$Cu^{2+} + 2e^- \rightarrow Cu. \quad [4]$$

Both simulations and experiments were performed on microstructured electrode arrays (MEAs) to provide a well-defined geometry and, therefore, a good basis to reliably compare individual experiments but also to use technically relevant geometries. Microelectrodes are very attractive in electrochemical applications due to their unique properties in comparison to macroscopic electrodes, e.g., higher current densities and faster establishment of the steady state. Consequences of changes in the electrolyte composition, in particular of the concentration of active species can be observed very quickly. Faster pulses in the applied potential/current can be used because of the lower capacitance of the electrical double layer. However, the small surface area of the microelectrodes leads to very small currents sometimes causing problems of low signal to noise ratio. One possibility to overcome this problem is the use of MEAs using parallel band electrodes. The overlap of the diffusion layers of adjacent electrode bands has

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to be considered and leads to a reduced mass flux to each individual electrode band in comparison to a single electrode. A lower current density or, in case of an electrochemical deposition, a lower growth rate is observed. As reported earlier the diffusion profiles can be distinguished roughly in three consecutive steps:[12,51] at the beginning of a reaction each electrode band can be treated separately, then the overlap has to be considered and finally the diffusion profile can be described as planar. Aoki et al. derived an equation for the boundary diffusion layer δ_N at single planar (without any elevation above the insulating surroundings) band electrodes (Eq. 5) which can be used to calculate the maximum current density j_N (Eq. 6).[52]

$$\delta_N = W_e/((\pi\theta)^{-1/2} + 0.97\text{-}1.1\exp[-9.9/\mathrm{abs}[\ln[12.37\ \theta]]]). \quad [5]$$

$$j_N = nFD\,c_0((\pi\theta)^{-1/2} + 0.97\text{-}1.1\exp[-9.9/\mathrm{abs}[\ln[12.37\ \theta]]])/W_e. \quad [6]$$

$\theta = D\,t/W_e^2$ serves as normalized dimensionless measure of the time t after start of the polarization using the width of the band W_e and the diffusion coefficient D of the rate-limiting reactant. Hongyuan et al. extended this approach to the case of multiple adjacent planar electrode bands by considering a shielding effect where the maximum current density j_{N2} becomes:[51]

$$j_{N2} = j_N(1\text{-}(2\arctan[(W_e^2/(W_e/\delta_N)^2 \text{-}((W_e + W_g)/2)^2)^{1/2}/(W_e + W_g)/2])/\pi). \quad [7]$$

The use of 7 or 6 is defined by the reaction time relative to t_c where the adjacent boundary diffusion layers begin to overlap, which can be calculated numerically by 5 with $(W_e+W_g)/2 = \delta_N$, with W_g representing the width of the insulating gap between the bands.

As a validity check we compared the calculated values for t_c based on this model to real-time and real-space simulations of electrodepositions of ZnO or Cu on MEAs with different sizes of W_e and W_g. Current densities were calculated based on 6 or 7 and compared to the FEM-simulated values. For this comparison we simulated the virtual case of a "deposition without deposition" (DWD)– which means a reaction consuming reactants but not forming any product. The electrochemical reaction limited by diffusion was simulated providing the resulting depletion of reactants in the electrolyte and the current density of the reaction without depositing material on the electrodes. For this case, therefore, no geometrical alteration, i. e., no remeshing was included in the simulations. As main contribution of this paper, however, simulations were performed including all changes brought forward by the deposition of material and, hence, changes of the electrode geometry were included, accommodated by appropriate remeshing steps. The simulated current densities compared well to experimentally observed values. Finally, the simulated shape of dendrites and the direct comparison with the experimentally observed dendrites was used for a detailed understanding of diffusion-controlled reactions.

Methods

Experiments. Microstructured electrode band arrays (MEA) of Au or Ag with band widths of 5 μm, 10 μm, 20 μm or 40 μm separated by a gap of same size and a height of 50 nm were prepared as described earlier.[12] The two sets of interdigitated metal bands were contacted by a wire and silver paste. Metal parts where no deposition should take place (contact pads, connections) were covered by an epoxy glue. Electrochemical deposition was performed by an IviumStat potentiostat in a typical three-electrode-setup with a Ag/AgCl-type reference electrode (RedRod, Radiometer Analytical, −40 mV vs. SCE) in a salt-bridge with saturated KCl solution. The ZnO depositions were performed at 70°C from a precursor solution 0.8 mM O_2 described earlier[12] at a potential of −1.1 V vs. SCE using a zinc-rod as counter electrode. Cu electrodeposition was performed at 25°C and −0.65 V vs. SCE using a 0.01 M $CuSO_4$ 0.1 Na_2SO_4 aqueous solution with a pH of 2 adjusted by H_2SO_4 according to[53] and a copper-rod as counter electrode, all chosen to ensure diffusion-controlled growth conditions. For comparison purposes, all current densities were determined by dividing the measured (or simulated) current by the original active area of the MEA. A changing electrode area during the time of deposition was explicitly considered in the supplemental material. The morphology of the deposited films was characterized by scanning electron microscopy (SEM) with a Zeiss Merlin and confocal laser microscopy with a Keyence VK-9700.

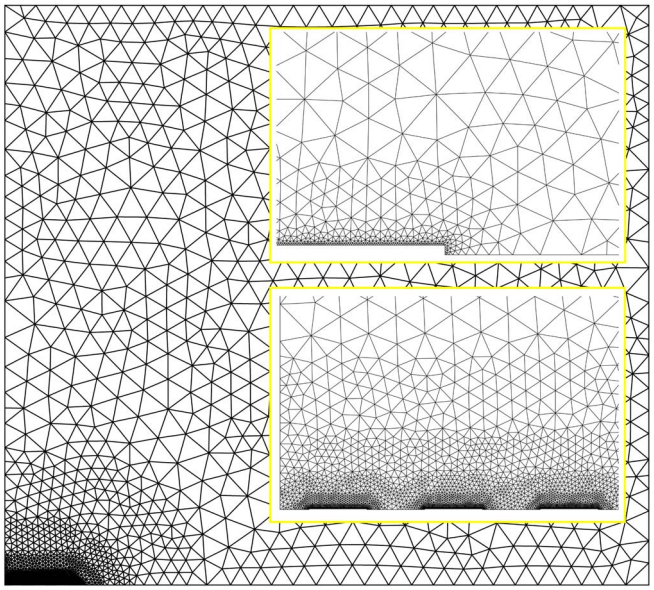


Figure 1. 2D mesh with increased refinement toward the 50 nm thick electrodes used for the Fluent solver. The main picture shows the whole simulation area for a 5 μm MEA with a height of 250 μm and a width of 272.5 μm. The bottom inset shows an enlarged view including all three bands. The top inset depicts details near the right edge of the central band.

Simulations. ANSYS CFX 14.0 and Fluent 16.2 with ICEM-CFD 14.0 as meshing tool were used to perform the simulations. The geometry used for the simulations is identical to the one used in the experiments if not otherwise noted. The simulations are 2D type in Fluent and pseudo 2D (single element 3D extrusion) type in CFX. By using a central symmetry plane of the structure only one half of the geometry had to be simulated. The simulation area has a height of 250 μm unless otherwise noted and a width of 250 μm measured from the outermost edge of the peripheral band in a given array. The mesh is chosen to be triangular or tetrahedron-based (in case of the pseudo 2D type) always using an increasing refinement toward the electrode bands (Figure 1). The simulation results reported stem from Fluent and were confirmed by CFX simulations which, however, did not provide relevant additional results.

To simulate electrochemical growth the mesh nodes and the boundaries of the electrodes are moving during the simulation according to the deposited material as a consequence of the simulated current in the preceding step. To compensate deformations of the mesh two ICEM-CFD based batch scripts were created: the first script smoothens the existing mesh to improve its quality and robustness ("smoothing") whereas the second script generates an electrode geometry out of this mesh, redefines the specifications for a new mesh and creates it ("remeshing"). The smoothing and remeshing times are determined by a macro during the execution, which monitors the minimum face area, the minimum edge length and the maximum cell skewness of the mesh. Limiting values for these parameters are manually adjusted in running simulations if the residuals become worse. Additionally, an interrupt of the simulation occurs if the smoothing or remeshing procedures fail a predefined number of times or because of numerical problems. In these cases, a manual remeshing is necessary. The implementation of these functions is organized by a custom user interface implemented in the Fluent main control window via a startup

Table I. Standard set of parameters and coefficients for the FEM simulations.

	ZnO (O_2)[56,57]	Cu (Cu^{2+})[53,57]
diffusion coefficent	$4.32 \cdot 10^{-9}$ m^2s^{-1}	$7 \cdot 10^{-10}$ m^2s^{-1}
bulk concentration	0.8 mol m^{-3}	10 mol m^{-3}
molar mass	$81.39 \cdot 10^{-3}$ kg mol^{-1}	$63.55 \cdot 10^{-3}$ kg mol^{-1}
Density	$5.67 \cdot 10^3$ kg m^{-3}	$8.96 \cdot 10^3$ kg m^{-3}

file. The diffusion equation in Fluent is established via a user-defined scalar transport equation modified by user-defined functions to establish Ficks law of diffusion and via a user-defined memory variable to store the gradient of the scalar for post-processing. In CFX a set of user-defined variables (one for each dimension) was used. The standard set of constants and coefficients for the simulations were summarized in Table I.

The simulations are solving Ficks law of diffusion on moving boundaries with $c = 0$ as boundary condition which represents the cathodes and the deposited material. In the gaps and at the bottom the flux of c was set to zero. As an approximation for a natural diffusion layer limit the value for the concentration of the diffusing species were set to the bulk concentration on the two boundaries diagonal to the MEA (top and right in Figure 1). The symmetry plane is placed in the center of the central gap for the 5 µm structures and in the center of the central band for the other sizes. Ficks law is used to reveal the diffusion-limited current density j_D on the surface of the cathodes:

$$j_D = nFD\nabla_S c \quad [8]$$

where n is the number of electrons, F is Faraday's constant, D is the diffusion coefficient and $\nabla_S c$ is the concentration gradient on the surface of the electrodes. Film growth is included in the simulations by calculating a mesh displacement Δd according to

$$\Delta d = zMD\nabla_S c\Delta t/\rho \quad [9]$$

with the number of electrons for the growth z, the molar mass M, the density ρ and the time step Δt.

Results and Discussion

FEM simulations were performed for the deposition of zinc oxide as a technically still challenging reaction and for the deposition of Cu as a well-established reference case. Before a comparison of simulations for MEAs with established models we validated our settings by simulations on single electrode bands. The simulated current density can be directly compared with Eq. 6 and, therefore, validate Eq. 5, too, which is important to distinguish the effects of overlapping diffusion layers from possible deviations in the basic description of individual bands. In Figure 2 the current densities for four band sizes and for the first 10 s are shown as calculated from Eq. 6 and as a result of the FEM simulations in black and green, respectively. Furthermore, for the 5 µm band width the current density for a completely embedded planar band is also shown in orange, to show that the deviations because of the electrode height, which is not included in the theory, are quite small. The current density for single band electrodes shows a time dependence expected for a diffusion-limited reaction. With increasing time, the depletion zone continues to spread in the electrolyte and the current density decreases until natural convection establishes a stable balance of depletion and ion exchange. The time scale of about 5 seconds needed to establish such steady state is consistently found and points at an approximate thickness of the diffusion layer ($\sqrt{D \cdot t}$) of about 150 µm. The comparison of theory and simulation shows a high degree of conformity for single bands and just with increasing time the curves start to differ because of our approximation of a maximum thickness of the diffusion layer given by natural convection. The shape of these curves is also characteristic for MEAs and is superimposed by the effects of an overlap of the diffusion layers of adjacent bands in Figures 3 and 4.

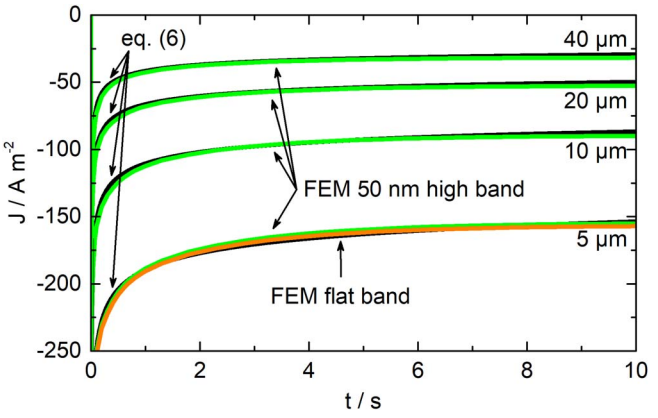


Figure 2. Current densities for 5 µm, 10 µm, 20 µm and 40 µm wide bands after Eq. 6 in black and for the FEM simulations (with a height of 50 nm) in green. The orange line corresponds to a simulation of an embedded (0 nm height) 5 µm wide structure. The data shows a very high degree of conformity for the established model for the current density at single bands and the present FEM simulations.

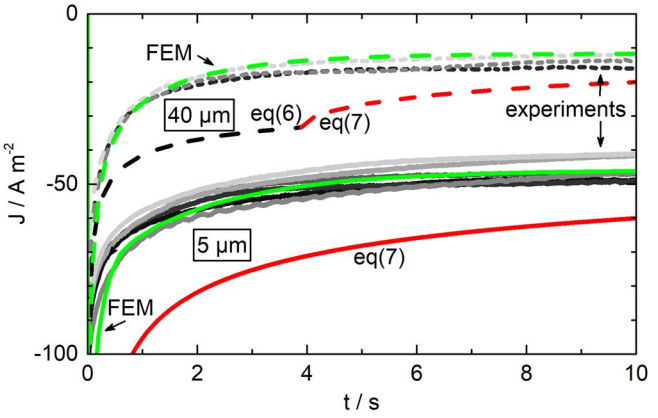


Figure 3. Current densities for the reduction of oxygen on a 5 µm (solid lines) and a 40 µm MEA (dashed lines) after Eqs. 6 and 7) in black and red, corresponding FEM simulations in green and experimental results in grayscale lines. The FEM data closely match the experimental results whereas the established models show substantial differences.

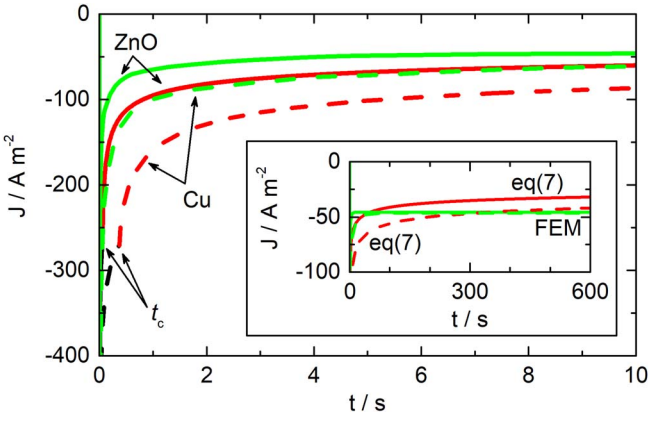


Figure 4. Current densities and the time t_c for DWD of porous zinc oxide (diffusion of O_2, solid lines) and copper (diffusion of Cu^{2+}, dashed lines). In black and red the current densities according to Eqs. 6 and 7 and in green for the FEM simulations (DWD) are shown. Eq. 7 leads to an infinite decrease of the absolute current density with time as revealed by the inset for 600 s whereas the simulations lead to a constant current density in the steady state in accordance with experiments.

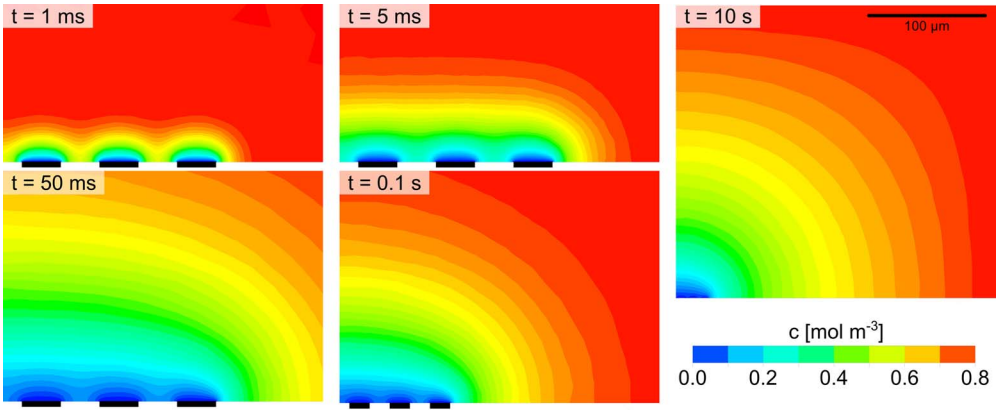


Figure 5. Time dependent concentration profiles for the oxygen reduction reaction (ZnO DWD) after Eq. 1 on a 5 µm MEA. The black bars represent the 5 µm wide electrodes (not their height) and the false color the concentration of oxygen. After 1 ms the diffusion layers already overlap but are almost symmetric to the MEA center. With ongoing time, the deep depletion region near the electrode surface shows asymmetry across the MEA. After 10 s a steady state has been established.

The zinc oxide system gives us the unique possibility to perform experiments where we can obtain the current density for the deposition without doing a deposition. This is realized by applying the potential in an electrolyte just without the addition of the zinc salt. In this way we conducted the oxygen reduction reaction (ORR) after Eq. 1 with the resulting current density which is also valid for Eq. 3 aside from a factor of 2. The experimental current density can therefore directly be compared to Eq. 7 and FEM simulations using a static mesh. The experiment was performed several times for a period of 10 s on a 5 µm and a 40 µm MEA and shown in Figure 3 in gray scale. The FEM data shown in green almost perfectly represent the experimental results, aside from an overestimation of the cathodic current density in the first milliseconds for the 5 µm MEA. The smaller current density in the latter experiment could relate to partly blocked electrodes by surface adsorbates, e.g., intermittently formed hydroxide ions from the ORR Eq. 1. The theoretical prediction by Eq. 6 and Eq. 7 in black and red overestimate the current density to a large degree. Due to the good correlation of simulations for single bands with Eq. 6 as seen in Figure 2 we assign the deviations for the MEA (Figure 3) to an underestimation of the influence of overlapping diffusion layers in the models.

From copper deposition, it is known that overlapping diffusion layers can have a significant impact on electrochemical growth. An asymmetric concentration profile can result in a differing rate of growth across an array, as directly shown in Ref. 46 by the concentration profile of a suppressor for copper growth. Therefore, as reported previously, an important value for the time-dependent electrochemical characteristics of a given system is the time t_c when adjacent diffusion layers start to overlap.[12] Until this time each band in the array can be treated separately but afterwards the depletion of the active species at one band electrode influences the current at adjacent electrodes. In an electrochemical deposition the growth rate will be influenced by the thickness of such depletion layers. In a previous study of ZnO electrodeposition,[12] values for t_c were calculated according to the theoretical model of Aoki et al.[52] Values for t_c can also be extracted from the present FEM simulations in real time and in real space by observing the time at which the depletion of the active species of two adjacent band electrodes start to overlap locally. This can easily be observed in graphical representations of the concentration as, e.g., shown in Figure 5 and Figure 6 for the ORR if the false color gradation of the simulated data is chosen sufficiently fine. We chose a gradation of 30 colors as an adequate compromise between a good differentiation by eye and acceptable precision for t_c. In this way, the relative concentration of the active species at the diffusion boundary layer amounts to 97% of the bulk value. This is sufficiently close to the quite common definition of 99%[54] and leads to just a small possible overestimation of t_c.

For a 5 µm MEA the simulation leads to a value of approximately 0.25 ms for t_c in the case of the ORR (ZnO) and of 1 ms for the case of Cu whereas with Eq. 5 we calculated a value of 58 ms for ZnO[12] and 360 ms for Cu. The period of time during which the electrode bands can be treated separately is simulated more than 200 times shorter as predicted by Eq. 5. This corresponds to the resulting current density as shown in Figure 3 and more detailed for a 5 µm MEA in Figure 4 for ZnO and Cu, where the resulting current densities calculated according to Eq. 6 (black line), Eq. 7 (red line) and from the FEM simulations (DWD, green line) are shown for the first 10 seconds and in the inset for longer times. The overestimation of the absolute current by Eq. 6 and Eq. 7 ranges from 55% at the unsteady transition from Eq. 6 to Eq. 7 in the beginning (at ca. 0.3 s) to 107% after 10 s and 222% after 600 s. A closer look on the time dependent concentration profiles reveals an origin for the lack of accuracy using Eq. 6 and Eq. 7. In Figure 5,

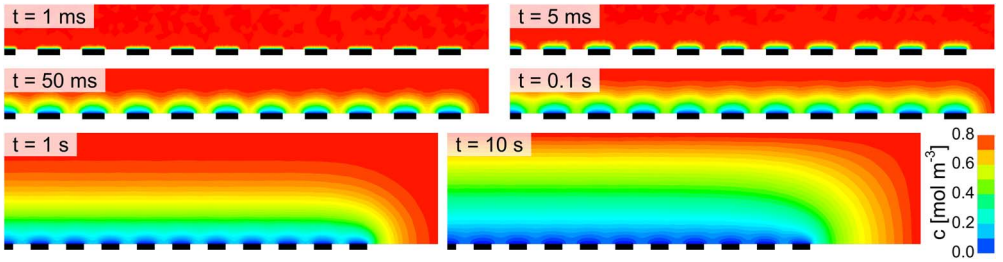


Figure 6. Time dependent concentration profiles for the oxygen reduction reaction as in Figure 5 but for a 40 µm MEA. The black bars again represent the 40 µm wide electrodes and the false color the concentration of oxygen. For 1 ms and 5 ms the boundary diffusion layers are not overlapping and show a square-shaped cross section with rounded edges. After 50 ms the diffusion layers overlap and show a semi-cyclic cross section. In the steady state after 10 s the deep depletion region is widely symmetric for the central bands but has an asymmetric shape on the peripheral bands.

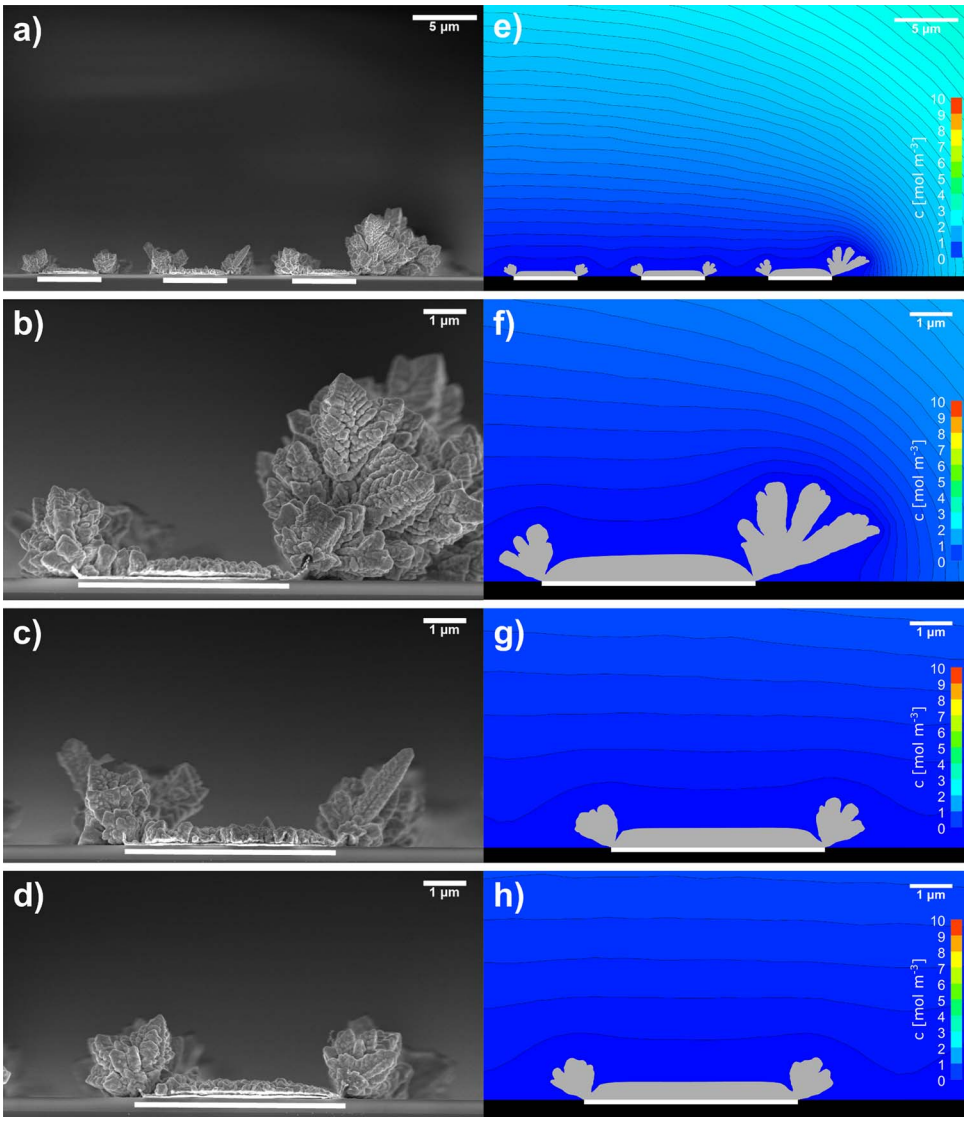


Figure 7. SEM (left) and FEM simulations (right) of deposited copper after 600 s of growth on a 5 µm MEA (electrode width marked by white bars) with equal length scales for associated images in a given row. Results are depicted for an overview of one half of the array (a, e), the peripheral band (b, f), the second band from the periphery (c, g) and one of the central bands (d, h). The false colors represent the concentration of Cu^{2+} ions. Aside from fluctuations of growth along a given band in the experiments as opposed to homogeneous growth in the simulations (see text) experiments and simulations match very well. The formation of dendrites and a growth rate increasing from the central part toward peripheral parts of the MEA are reproduced well.

the depletion of oxygen is shown in a false color representation as a result of an FEM simulation of a 5 µm MEA. Close to the electrode bands the concentration profile appears symmetrical and semi cyclic in cross section for each band in an early stage (1 ms) of the depletion of oxygen and widely identical for each band in the array. For longer polarization times, however, the concentration distribution is no longer symmetrical on each individual electrode band nor identical across the MEA, especially well represented at 50 ms and 0.1 s. Furthermore, for a FEM simulation of a 40 µm MEA even the concentration profile in the early state does not show a semi-cyclic cross section (Figure 6, 1 ms and 5 ms). A shielding effect based on a semicircle as assumed in Eq. 6 and Eq. 7, therefore, reflects the conditions of depletion on a MEA just in a small period of time, which strongly depends on the width of the bands. Correspondingly, for oxygen depletion at a 10 µm MEA the deviations for t_c calculated by Eq. 5 with 181 ms and approximately 700 µs for the FEM simulation and at a 40 µm MEA with t_c calculated by Eq. 5 with 3.7 s and approximately 15 ms for the FEM simulation are even higher than for the 5 µm MEA. Simulations performed for Cu DWD led to very comparable results with slightly shifted times (see above for t_c). It should also be noted (Figure 4) that the FEM simulations in contrast to Eq. 7 lead to a constant steady-state value of the current densities after about 50 s in accordance with the experimental results due to the limitation of the depletion layer to the size of a natural convection layer.

The current density during an electrodeposition is not only influenced by reactant depletion but by geometrical changes of the electrode by the deposited material itself. Such growth is explicitly simulated by a deforming mesh according to Eq. 9 for the electrodeposition of both porous ZnO and compact Cu. The comparison with experimental data, however, is focused on Cu since very reliable and reproducible experimental results could be obtained in this system whereas the electrodeposition of ZnO turned out to be very sensitive to subtle fluctuations (see also below). For such comparison we chose a simulated and an electrochemical deposition time of 600 s to ensure that the steady state was reached and to include the results from all earlier times. In Figure 7 the morphologies resulting from this growth simulation and from an electrochemical deposition of Cu on a 5 µm MEA are shown. The resulting morphologies of the

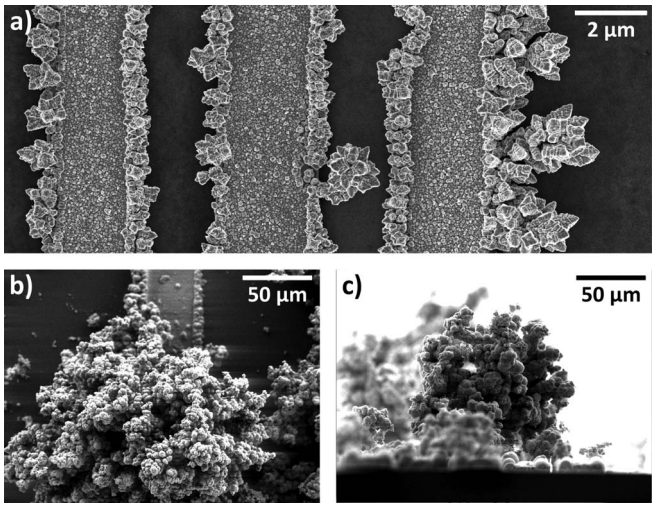


Figure 8. (a) SEM top view of 3 electrode bands of a 5 µm MEA (center of MEA to the left) after a copper deposition for 600 s. Dendrites are seen on the edges whereas in the center of each band a film of homogeneous thickness is formed. The dendrites have a preferred growth toward the gap between to bands and toward the periphery of the array. SEM analysis of an electrodeposited film of ZnO in top view (b) and cross section view (c) are shown for direct comparison to show the morphological similarity at, however, strongly enhanced dendrite formation. The films were obtained under rotation of the sample and, hence, reduced thickness of the diffusion layer as well as some structural differences of the MEA described in Ref. 12.

SEM cross-sections (a to d) and from the simulation (e to h) show very high similarity and provide further proof for the quality of the simulation. In a) and e) an overview of three bands in the array is shown. The increase in amount of deposited material and the increase in dendrite size from the bands in the center of the MEA (left side) toward the peripheral bands of the MEA (right side) observed in the experiments is very well reproduced by the simulations. In b) to d) (SEM) and f) to h) (FEM simulation) images at increased magnification of the individual bands are shown. The simulations very well reproduce the cross section of deposited Cu consisting of a widely homogeneous film thickness in the center of each band which becomes smaller toward each band edge where a significantly increased amount of deposited Cu (dendrite growth) was observed in the experiments (see also Figure 8a). The experiments, further, show that the shape of the enhanced copper growth on the edges develops from a more cylindrical (widely circular cross section) enhancement in the center to dendrite-like structures at the peripheral band edge. These same characteristics are observed in the images from the simulation. The specific leaf-shaped morphology of the Cu dendrites cannot be reproduced by the simulations since crystallization kinetics along different crystal directions[55] were not considered in the model. Further, the growth rate of the dendrites seems to be underestimated in the FEM simulations relative to the growth rate on the planar areas in the center of the bands. Several reasons can be discussed to cause such deviation. The most important factor will be the third dimension (along the bands) which is of influence in the experiments (Figure 8a) but is not considered in the simulation. Every dendrite along a band edge has a certain distance to the next one which can be related to the overlapping diffusion layers around individual growing dendrites similar to the characteristics modeled in the 2D characteristics comparing the different edges. Between dendrites, depletion suppresses further growth, close to a dendrite the higher reactant concentration enhances its subsequent growth. Initial dendrite formation can be triggered by morphological imperfections (Figure 9) which might also be present in the metal edges as a result of the lift-off during photolithography. Further, as already mentioned in the discussion of the observed specific shape of the dendrites, crystallization kinetics are not considered in the simulation. Faster growth in a given crystallographic direction, however, could lead to an increased dendrite growth rate along the steepest concentration gradient and, hence, lead to faster dendrite growth than modeled by a homogeneous constant growth rate. Finally, the diffusion coefficient used for the simulations may not perfectly match the chosen composition of the deposition bath although the best estimate for the diffusion coefficient and the corresponding procedure of Quickenden and Jiang[53] was used. However, different literature values for the diffusion coefficient of copper ions in water at room temperature are listed and it is emphasized that the diffusion coefficient can vary considerably even in formally equal solutions.

For the electrodeposition of ZnO, a similar growth pattern as discussed above for Cu (homogeneous planar films in the center of an electrode band, dendrite formation at the edges of bands, increasing in size from central bands toward peripheral bands of a given MEA) was found earlier[12] and reproduced in the present experiments (Figures 8b, 8c). When compared to the deposition of Cu, however, it is seen that dendrite formation occurred to a significantly higher extent in the case of ZnO. Large strongly branched dendrites were formed on the edges of electrode bands but were even found in the central part of electrode bands where a planar film is typically formed. These similarities and differences were well reproduced by the corresponding FEM simulations (Figure 9). Small protrusions with a height of 50 nm were randomly placed onto the electrode bands to simulate subtle fluctuations in electrode morphology which in the experiments would stem from irregularities in the Au electrode bands of a given MEA or from local fluctuations in the deposition rate. Figure 9 shows the growth during the deposition of Cu (left) and ZnO (right) on the second electrode band from the periphery of a 5 µm MEA for different times under consideration of a protrusion of 50 nm height and 88.4 nm width positioned 1.314 µm from the left edge of the band. It is revealed that the relative temporal development of the depletion zone of the respective reactant proceeds quite similar in the two depositions studied. After 0.1 s the boundary diffusion layer has grown a bit larger for the porous ZnO deposition but qualitatively did not differ significantly from the corresponding Cu case. This still holds after 10 s of simulated deposition but different from Cu the coverage by porous ZnO can be detected on the chosen scale. This is consistent with an about 4-fold higher volume growth rate of ZnO that can be estimated from the stoichiometric factors (Eq. 3 and Eq. 4), the molar masses, densities, reactant concentrations and diffusion coefficients assuming complete conversion under diffusion limitation as assumed in the simulations. At this stage a closer look to the protrusion reveals that for the ZnO system a small influence in the concentration profile up to 0.2% can be observed whereas in the Cu system this is not the case. Aside from the higher amount of porous ZnO deposited after 100 s and pronounced dendrite formation at the edges of the electrode band for both materials, a difference in film growth at the protrusion can be observed. Whereas the edges of the protrusion act as sites of Cu dendrite growth the whole protrusion acts as site of increased growth in the ZnO system and the formed porous ZnO dendrite significantly influences the surrounding concentration profile unlike the Cu system in which a rather homogeneous profile is established across the protrusion site. After 300 s the porous ZnO dendrite at the protrusion site has reached a significant beyond the height level of the deposited film leading to a severely distorted concentration profile of the reactant as opposed to the Cu at the protrusion site which has grown to a height comparable to the surrounding planar part of the deposit on the electrode band leading to an undisturbed concentration profile across the protrusion site. Finally, after 600 s the porous ZnO dendrite has grown to a size and shape close to the dendrites at the edges of the band as opposed to the copper deposit on the protrusion site which is lying within the planar copper area covering the band. The original decreased growth of ZnO and Cu at both ends of the protrusion caused by increased reactant depletion is preserved throughout the deposition. No further amplification, however, of Cu growth at the protrusion site has occurred and can be expected with ongoing deposition. The observation (Figure 8) of significantly more pronounced dendrite formation at the edges of bands for ZnO compared to Cu as well as the spontaneous dendrite formation on the electrode band which was

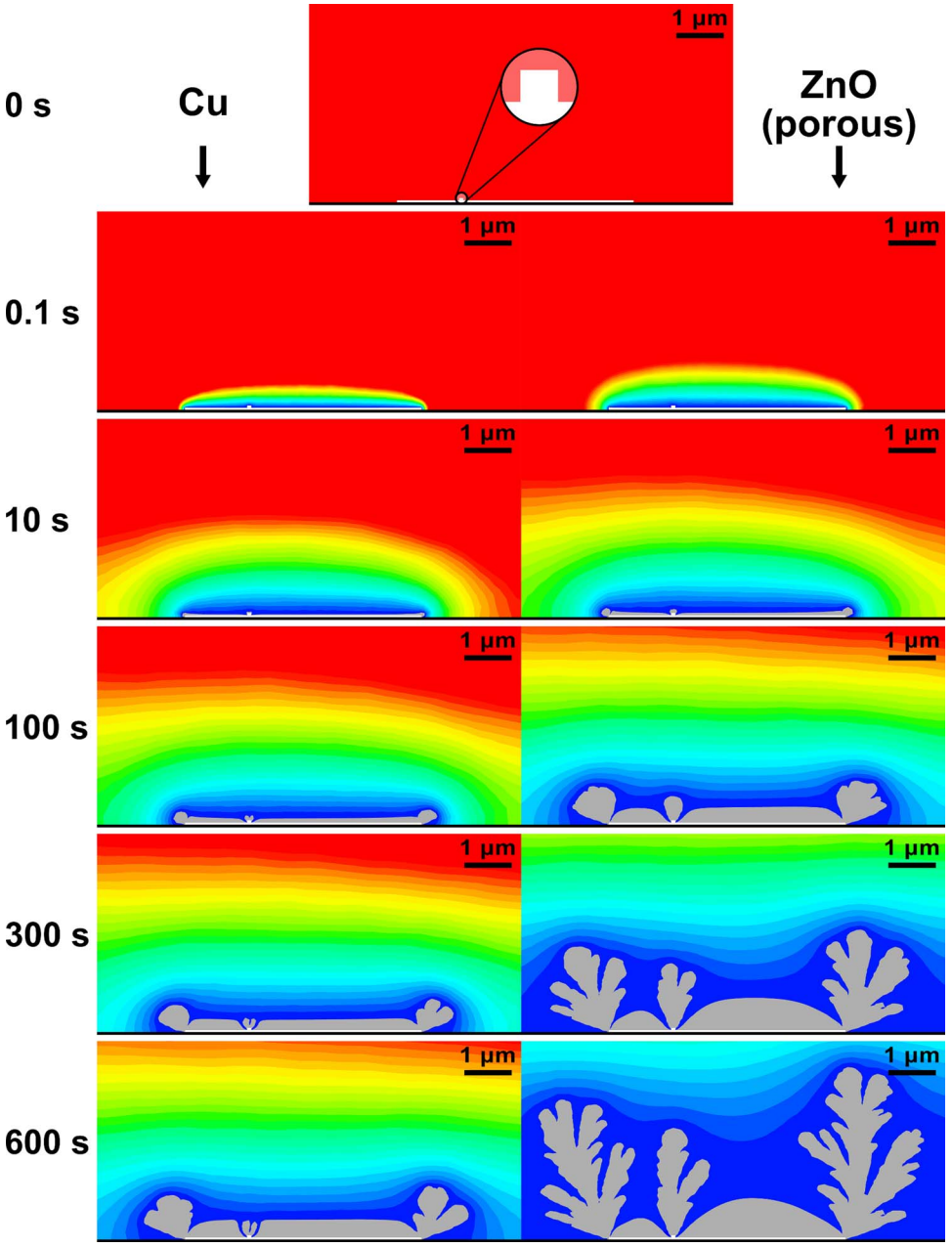


Figure 9. FEM simulation results for different times during the growth of Cu (left) and porous ZnO (right) on a 5 µm MEA with a randomly placed protrusion (50 nm × 88.4 nm in cross section; enlarged view in image at 0 s) of the electrode material. The reactant concentration (Cu^{2+} or O_2, respectively) is depicted in false color ranging from 5 (or more) % of the bulk concentration in red down in 25 color steps toward 0.2 (or less) % in dark blue. The protrusion gets strongly enhanced during growth of porous zinc oxide but not for the deposition of copper.

observed for ZnO but not for Cu are both perfectly explained by the simulations as arising from the relative volume growth rates in balance with reactant diffusion along the established concentration profiles (see Supplemental Material also) without consideration of specific differences in crystallization kinetics.

Conclusions and Outlook

Dendrite formation was simulated by solving Fick's law of diffusion considering mesh displacement. This reveals that even fractal-like structures can be formed in an electrochemical deposition caused by limitation of mass transport by diffusion toward the electrodes. The FEM simulations of the diffusion layer in an electrochemical deposition reaction on MEAs which are in very good agreement with the experimental results show that the overlap of adjacent diffusion layers occurs at considerably shorter reaction times than predicted by an existing model.[51] Application of the simulations, therefore, leads to increased precision in the calculation of the resulting current densities and to correct estimations of the influence of overlapping diffusion layers in a given electrochemical deposition. Further, the simulations correctly yield a constant deposition current in the steady state as experimentally observed.

It could be shown that an asymmetric diffusion profile establishes over the whole electrode array and across individual bands especially in the periphery of a given array. Such asymmetry perfectly explains the enhanced deposition of material on the edges of individual bands, to an increasing extent toward the peripheral bands and that the largest dendrites are consistently found on the outermost edge of

the peripheral band. The morphology of films deposited under diffusion control is thereby qualitatively and even quantitatively explained by the present simulations using frequent remeshing procedures under conditions of a permanently changing electrode geometry on length scales of up to 10 μm. A special case of an intrinsic smoothing behavior with suppressed dendrite formation at morphological fluctuations of a given electrode as often desired in technical processes was detected and explained by the established concentration profile for the electrodeposition of Cu under the present deposition conditions. Such smoothing, however, could be experimentally established on widely planar electrode areas only but not along the more critical edges of a given electrode band. Additional simulation work is, therefore, needed to define the range of protrusion size relative to surrounding electrode homogeneity for which such smoothing can be established, also depending (see also supplemental material) on the volume growth rate relative to the rate of diffusion.

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